Article

Effectiveness of Temporary Portable Rumble Strips in Utah

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Abstract: Transportation agencies regularly use Temporary portable rumble strips (TPRS) to alert drivers to upcoming work zones or other changing traffic conditions, based on previous research showing TPRS to effectively reduce traffic speed and increase driver attentiveness. However, questions persist regarding the continued effectiveness of TPRS as drivers become accustomed to them, the preferred spacing of TPRS, and the need for workers to periodically enter the roadway and adjust the TPRS when they become displaced. This research presents the results of a field experiment at four sites in Utah with varying deployment conditions and TPRS spacings. The sites were instrumented with radar speed detectors and video cameras to capture change in speed, braking behavior, avoidance, strip displacement, and worker exposure. Results indicate that the effect of TPRS on speed reduction is inconclusive. Drivers who braked in response to TPRS typically did so before reaching the devices, suggesting that visual cues may be as influential as tactile feedback for most drivers. Avoidance behavior was common among motorcycles and occurred occasionally among other vehicles. Driver responses were more strongly influenced by site-specific factors—such as roadway geometry and sight distance—than by TPRS spacing. TPRS displacement occurred gradually under traffic but was not clearly related to traffic volume or speed, suggesting that device condition play a larger role. Maintenance requires traffic gaps of approximately 34 seconds, which may not be feasible at higher-volume sites. These findings highlight the need for further evaluation of TPRS effectiveness, standards for device condition, consideration of alternative measures for long-term work zones, and increased contractor discretion based on site conditions.

Keywords: Temporary Portable Rumble Strips; Work Zone Safety; Driver Attention

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1. Introduction

Temporary portable rumble strips (TPRS) have been used since at least the early 2000's to alert drivers to upcoming work zones. Early adhesive strips have been largely replaced by non-adhesive steel and rubber pads heavy enough to resist movement but light enough to be handled by workers [1–3]. Prior research has generally shown that TPRS reduce traffic speed and increase driver attentiveness [4,5], with the effect being most pronounced near the first set of strips and when drivers can see a flagger or additional barriers [6]. There is also some evidence that TPRS reduce crash rates [7]. These findings have led to the wide adoption of TPRS in work zones across many states.

In spite of their common use, there are several unanswered questions related to TPRS deployments. First, some studies on the effectiveness of TPRS have been conducted on closed courses with instrumented vehicles [8,9]; it is unknown whether this is a generalizable finding for work zones with open-road conditions. Similarly, as drivers become accustomed to TPRS, they may become desensitized to them or avoid them [3], making them less effective safety measures. Second, a review of state specifications for TPRS spacing shows that there is little evidence to support standards related to the spacing of strips within a TPRS array, and that desired specifications vary widely across states. Some states, such as

Table 1. Data Collection Sites

Site	Roadway	Location	Collection Dates Start	End	Speed Limit	Work Zone Limit	Closure Type
1	SR-12	Mile marker 6.5	2025-07-07	2025-07-11	55	35	Full lane
2	US-6	20 miles East of Wellington	2025-07-15	2025-07-18	65	Not recorded	Full lane
3	I-70	Clear Creek Summit	2025-07-29	2025-08-01	80	60	Passing lane
4	US-191	North of summit	2025-08-05	2025-08-08	60	40	Full lane

Wisconsin, Florida, and Colorado, specify a single spacing independent of speed [e.g., 10]. Some states such as Maryland, Virginia, Minnesota, and Utah vary spacing from 10 to 20 ft. based on the posted speed limit [e.g., 11]. Finally, TPRS can become displaced with traffic, limiting their effectiveness or creating a potential hazard for drivers [4,9,12]. Workers must therefore periodically enter the roadway to install the strips and adjust them when they become displaced.

This paper reports a field experiment to evaluate driver reaction to TPRS and observe strip displacement under traffic. Four sites in Utah were instrumented with cameras and speed-detecting radar units, allowing researchers to evaluate change in speed, braking behavior, avoidance, displacement, and worker exposure. Four TPRS spacings were applied to each site, including a no-TPRS control. This paper is organized as follows: A methodology section describes the experimental design and methods used to collect data on driver behavior and TPRS displacement. A subsequent results section presents the results of the experiment, followed by a discussion of the findings with associated limitations of this research. A final concluding section discusses the implications of the findings for practice and future research.

2. Methodology

This section describes the methods and procedures used to select sites and collect data on driver behavior and TPRS displacement. More details on the reasoning behind the methodological decisions are available in an associated research report [13].

2.1. Site Selection

The research team identified four sites at UDOT work zones in the Summer 2025 construction season compatible with the goals of this research. These sites are at the locations shown in Figure 1 and data collection dates described in Table 1. Site selection criteria included sequencing of researcher travel, planned construction schedules, variety of site speeds and roadway types, and expected traffic volume. Nearby traffic counting stations provided volume data to confirm that sufficient samples to conduct statistical analysis were available.

The research team decided on four layouts for the arrays of TPRS to observe the impact of TPRS on driver behavior and strip displacement. The layouts are as follows:

1. **No TPRS**: TPRS are not installed.
2. **UDOT**: The current recommended specifications in Region 4, which are 10 ft. for roads with a posted speed limit up to 45 mph, and 20 ft. for roads with a posted speed limit of 50 mph or higher.
3. **1:2**: TPRS spaced in feet at half the value of the posted speed limit in miles per hour (e.g. a 60 mph road will have spacing of 30 ft.).
4. - **Long**: TPRS spaced between five to twenty ft. longer than the 1:2 spacing specification, depending on the posted speed limit.

Figure 2 illustrates the overall instrumentation plan applied at each site. The instrumentation plan consists of two video cameras, Camera 1 and Camera 2, located upstream of the first TPRS array and at the TPRS array, respectively. There are also two trailer-mounted radar speed detection devices manufactured by Wavetronix. These are called Radar 1 and Radar 2 and are located at the first TPRS array and downstream of the first TPRS array,

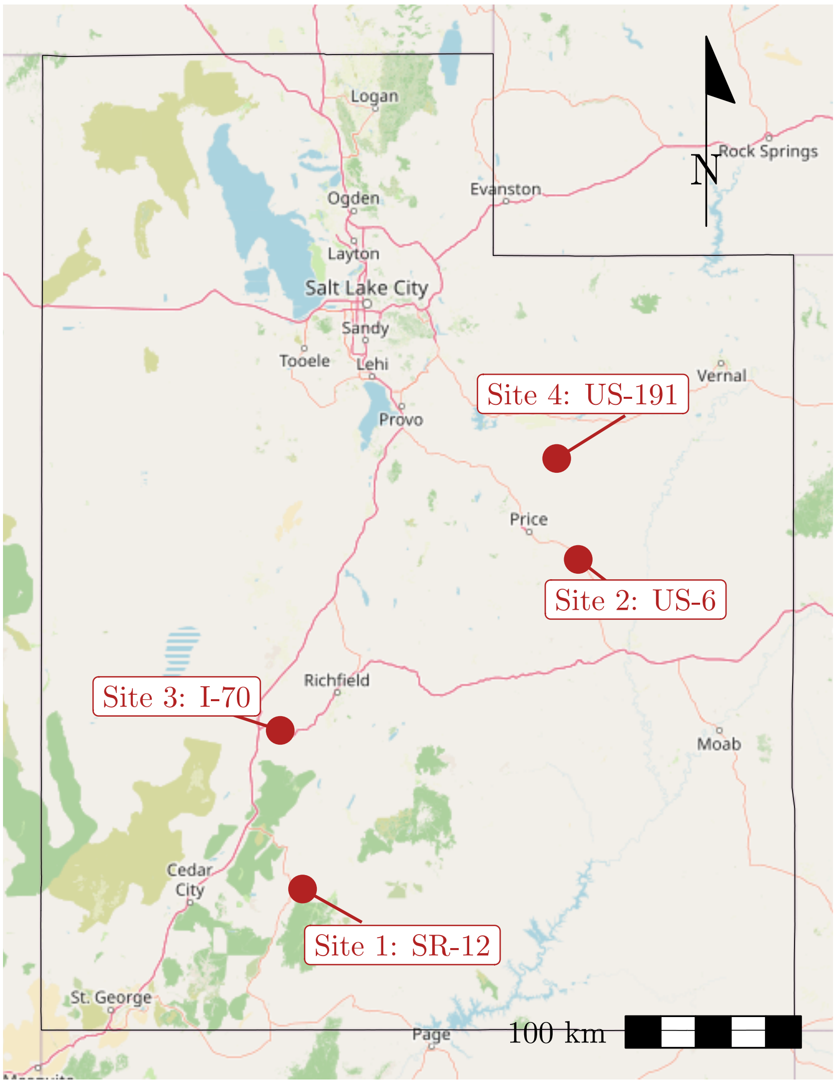


Figure 1. Map of Utah with study locations.

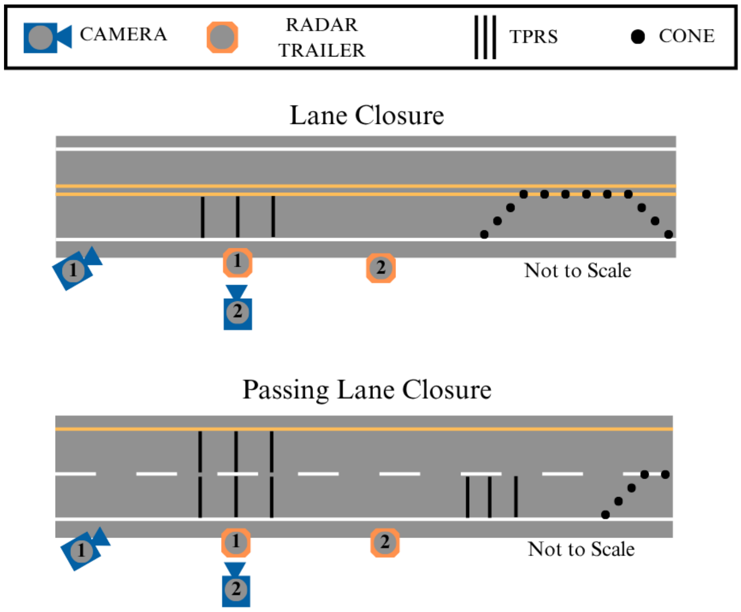


Figure 2. Instrumentation plan at lane closure and passing lane closure.

Table 2. Separation [ft] Between Radar Units

Site	Roadway	UDOT	NO TPRS	1:2	LONG
1	SR-12	762/875	851	878	898
2	US-6	516	516	516	516
3	I-70	800	804	782	765
4	US-191	782	748	781	759

respectively. Camera 1 and Radar 2 were each installed where space on the shoulder of the roadway was available. Table 2 provides the actual distance separating Radar 1 and Radar 2 at each site on each day. Observations at SR-12 were divided into two days, resulting in two different measures.

The research team placed the quantifiable metrics of this study into two basic categories. The first is driver behavior as observed through change in speed, braking response, and avoidance behavior. The second consists of measured quantities related to TPRS movement, including individual strip displacement and worker exposure during adjustment. The subsections below describe the setup used in this observational study to measure these variables.

2.2. Driver Behavior - Change in Speed, Braking, and Lane Departures

TPRS are designed principally to gain and increase driver attentiveness and not primarily to reduce speeds; however, measuring driver attentiveness in the general population at an active worksite presents technical and safety challenges. Change in speed, braking response, and TPRS avoidance have been used in previous studies as appropriate proxy measures [3,4,14]

Wavetronix units use radar to measure the speed of vehicles within their area of effect based on programmed lane location information adjusted for each site on each day. Accompanying software provides vehicle counts, average speeds, and 85th percentile speed measurements for the vehicles passing the units in bins specified by the user; this study used 15-minute bins. By taking the difference in the 85th percentile speed captured in the same 15-minute bin between Radar 1 and Radar 2, it is possible to measure the change in 85th percentile speed.

Camera 1 was placed roughly 100 ft. upstream from the first TPRS array and provided a view of the rear of vehicles as they encountered the strips. The research team watched the Camera 1 video and marked vehicle classification and relevant behavior using software developed by the research team for this purpose. The researchers recorded when each vehicle crossed the TPRS; classified each vehicle as either a motorcycle, passenger vehicle, or truck; identified whether a vehicle illuminated its brake lights before or after encountering TPRS; and identified whether the vehicle avoided the strips.

The researchers classified each vehicle traversing TPRS. Vehicles with fewer than four wheels were classified as motorcycles. Vehicles with two axles and four wheels were classified as passenger vehicles. Vehicles with more than four wheels or more than two axles (exclusive of trailers) were classified as trucks. This aligns with the FHWA's 13 vehicle category classification with Class 1 as motorcycles, Classes 2 and 3 as passenger vehicles, and Classes 4 through 13 as trucks [15], and is consistent with the definition of truck used in calculating annual average daily truck traffic [16].

The research team classified vehicles as braking before TPRS, after TPRS, or not braking at all. Vehicles driving on the shoulder or crossing the center dividing line ahead of TPRS — either partially or completely — were identified as avoiding the TPRS. Vehicles engaged in passing the vehicle in front of them were not considered to be avoiding the strips.

2.3. TPRS Displacement and Worker Exposure

Camera 2 was mounted to the same point as Radar 1 and had a top-down view of the TPRS array, centered on the middle strip of the array. This perspective allowed the team to see how much the center strip of the TPRS array was displaced over time and how much time workers spent adjusting the TPRS.

The research team used a transparent overlay to track TPRS displacement over time. The overlay was based on PSS standards as outlined in their product guide, namely that the strips can move up to 5 ft. forward or back, 2 ft. to either side, and can rotate up to a 3 ft. differential between ends or roughly 15° [17]. To provide more nuanced description of displacement, the researchers developed five categories of displacement based on these overall specifications: reset (no movement), some, moderate, significant, and out of specification. Figure 3 shows the overlay with dimensions for each area marked. The overlay was warped to approximate the perspective of the camera as shown in Figure 4.

To quantify and normalize the vehicle impact force to displace a strip, the research team estimated the cumulative momentum — speed multiplied by mass — applied to TRPS. The weight of each vehicle was assumed based on the vehicle classification and a mean class weight derived from national data [18]. Each vehicle's speed was assumed to be the mean speed of its 15-minute time bin as measured by Radar 1.

Camera 2 footage allowed researchers to evaluate worker exposure to traffic when correcting the strip alignment. The research team recorded when workers arrived on site to move TPRS, when vehicles passed while workers were on site, when workers started looking for a gap in traffic, when workers entered the road to adjust TPRS and exited the road, and when the workers left the site. The time when workers were actively engaged in seeking to enter the road was subject to a probabilistic critical gap analysis [19] to identify the length of headway workers will accept for this task.

3. Results

This section describes the research results with regards to driver speed reduction, braking behavior, and TPRS avoidance, followed by the results related to TPRS displacement and related worker exposure.

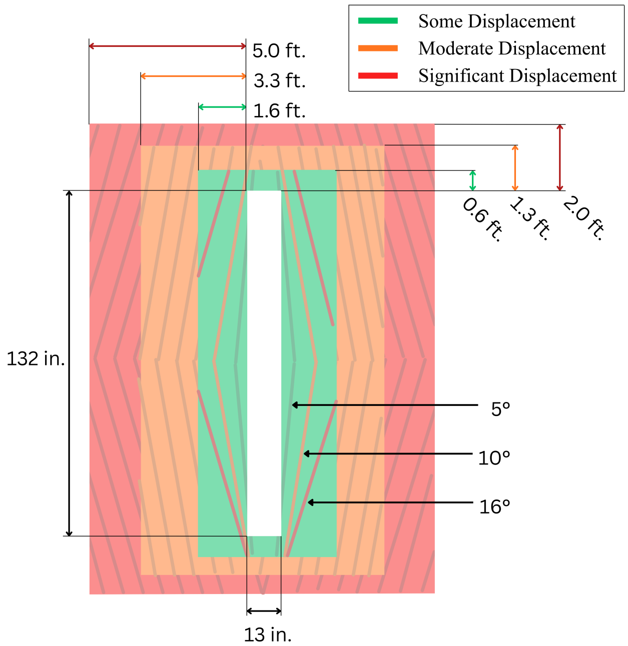


Figure 3. TPRS displacement overlay.

3.1. Speeds

The Radar devices record the mean and 85th percentile speeds for vehicles passing the detectors aggregated to 15-minute bins, meaning that in an 8-hour observation period, 48 85th-percentile speeds are recorded at each detector. Figure 5 shows the difference in mean 85th percentile speeds between Radar 1 and Radar 2 and the 95% confidence interval of this difference. A positive difference implies that the 85th percentile speed *increased* between Radar 1 and Radar 2. The results of this analysis do not show any clear correlation between change in speed and TPRS spacing, though in many conditions there is a reduction in speed. If the TPRS spacing had a consistent impact on speed, the relative position between the spacing types would be consistent between sites. The data do not show this pattern; rather, the impact of TPRS presence or spacing varies inconsistently between sites. Notably, there are cases on US-191 where vehicle speeds increased after drivers traversed the array of TPRS. Although this might seem to contradict the current consensus found in the literature, some studies did acknowledge that TPRS influence on speed is highly localized [20,21]

3.2. Braking Behavior

Using Camera 1, the research team tracked drivers' braking response to TPRS as another proxy measure of driver awareness. Table 3 shows a multinomial logit model of braking behavior independently estimated for each site. In this model, significant positive numbers are correlated with an increase in the associated behavior relative to the not-braking reference. Vehicle type is not consistently and significantly correlated with braking behavior. Motorcycles and trucks are both more likely than passenger cars to brake before the strips at I-70, trucks more likely at US-191, and motorcycles at US-6. There is no correlation between vehicle type and after-strip braking at any site. TPRS spacing is also not consistently and significantly correlated with braking behavior. Relative to the no-TPRS condition, before-strip braking is more likely at SR-12, US-6, and I-70 at all spacing types. Before-strip braking at US-191 is oddly somewhat less likely with strips installed than without strips in place. After-strip braking is more likely with strips installed on I-70, and with UDOT spacing on US-6. But after-strip braking is less common with strips installed on SR-12 than without strips installed. The fact that many vehicles brake before TPRS may suggest that the influence of TPRS is primarily visual rather than tactile for most road users. If drivers are reacting to TPRS on sight rather than striking the strips, this could explain

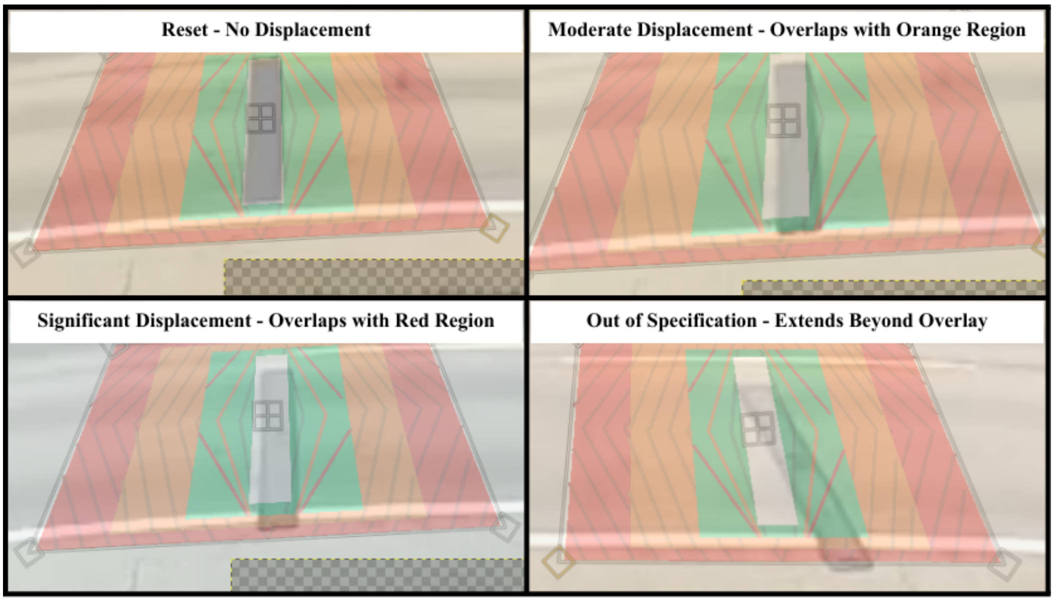


Figure 4. TPRS displacement examples.

why motorcycle drivers seemed to brake more when TPRS are not present. Motorcycle drivers may be braking before they enter the camera's field of view when they see the TPRS installed, and thus not being counted. But when TPRS are not installed, motorcycle drivers may be braking later in response to seeing the lane closure rather than the TPRS. This interpretation is speculative and more research is needed.

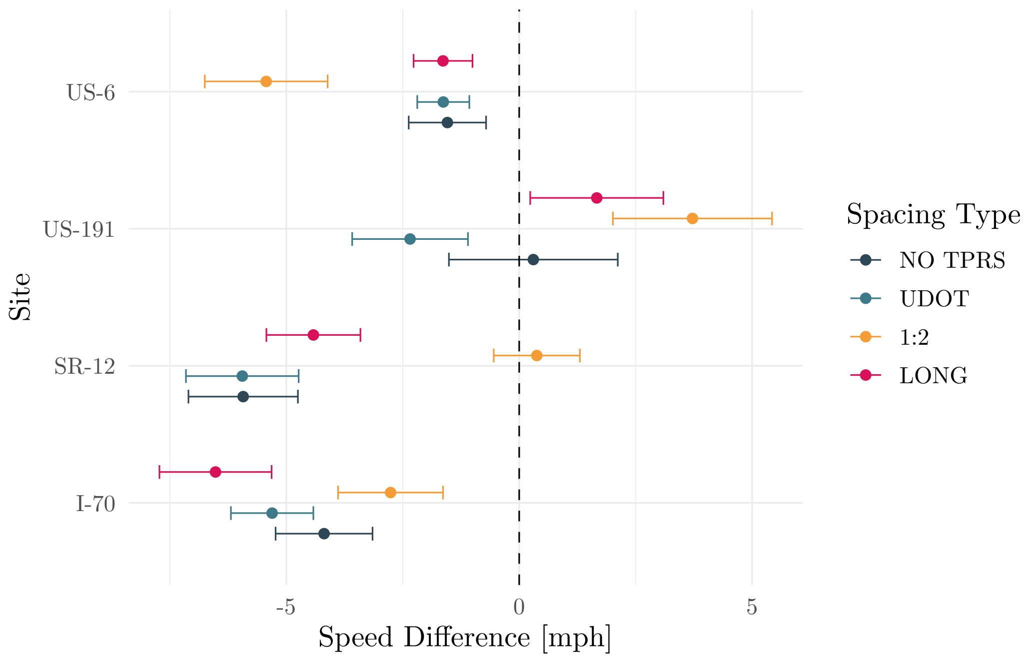


Figure 5. Change in 85th percentile speed between radar speed units with 95 percent confidence interval on difference in speeds.

Table 3. Multinomial Models of TPRS Braking

	SR-12		US-6		I-70		US-191	
	Before	After	Before	After	Before	After	Before	After
(Intercept)	-2.929***	-3.378***	-4.064***	-5.292***	-3.730***	-7.336***	-2.019***	-5.367***
	(0.125)	(0.161)	(0.201)	(0.387)	(0.142)	(1.008)	(0.152)	(1.026)
Trucks	0.033	-0.580	0.021	-0.350	1.490***	0.333	1.631***	-1.088
	(0.157)	(0.597)	(0.139)	(0.384)	(0.094)	(0.367)	(0.136)	(1.114)
Motorcycles	0.435	-7.051	1.338**	1.322	1.860**	-7.554	0.669+	-6.144
	(0.302)	(28.714)	(0.473)	(1.034)	(0.699)	(179.374)	(0.387)	(45.540)
UDOT Spacing	1.403***	-0.771*	1.830***	1.164**	0.554**	2.296*	-0.339*	-4.771
	(0.151)	(0.343)	(0.213)	(0.426)	(0.174)	(1.062)	(0.167)	(10.387)
1:2 Spacing	1.155***	-1.683***	1.106***	1.066*	1.290***	2.321*	-0.517**	0.443
	(0.145)	(0.410)	(0.254)	(0.486)	(0.151)	(1.050)	(0.166)	(1.225)
Long Spacing	1.161***	-1.495***	1.408***	0.485	1.315***	3.157**	0.054	0.680
	(0.147)	(0.389)	(0.218)	(0.465)	(0.154)	(1.024)	(0.164)	(1.225)
Num.Obs.	4963		6324		5933		1796	
RMSE	0.282		0.209		0.238		0.328	
McFadden ρ^2	0.036		0.036		0.089		0.096	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

3.3. Avoidance Behavior

The research team observed lane departures where vehicles appeared to avoid TPRS. It is the opinion of the researchers who watched the camera footage that most vehicles that departed their lane to avoid TPRS did so in an orderly manner similar to passing another vehicle, as opposed to a chaotic adjustment. There was no differentiation in the observations between partial lane departures, where only one side of tires crossed the center line or right-side line, and full lane departures, where all tires crossed either edge of the lane. On I-70, where TPRS were laid across both lanes, there were approximately 10 vehicles that drove fully in the shoulder to avoid the TPRS, but these were not classified separately.

Table 4 presents a series of binomial logit models estimating the probability of lane departure at each site. In this model, significant positive coefficients indicate that vehicles are more likely to avoid TPRS. Motorcycles have a consistent and dramatically higher rate of lane departure than either passenger vehicles or trucks, but trucks and passenger vehicles avoid at the same rate. This finding confirms prior literature that motorcycles, as vulnerable road users and more agile vehicles, are both motivated and capable of avoiding the TPRS. Although some passenger vehicles and trucks do leave their lane to avoid the TPRS, the vast majority do not. The significance of the model coefficients showing TPRS avoidance at most TPRS layouts is a feature of TPRS avoidance being impossible when they are not deployed, which is the reference alternative in the model, and not a finding of importance. What is more important is that the estimated coefficients associated with TPRS spacing are not significantly different from each other at each site, indicating that the spacing does not incite or discourage TPRS avoidance.

3.4. Strip Displacement

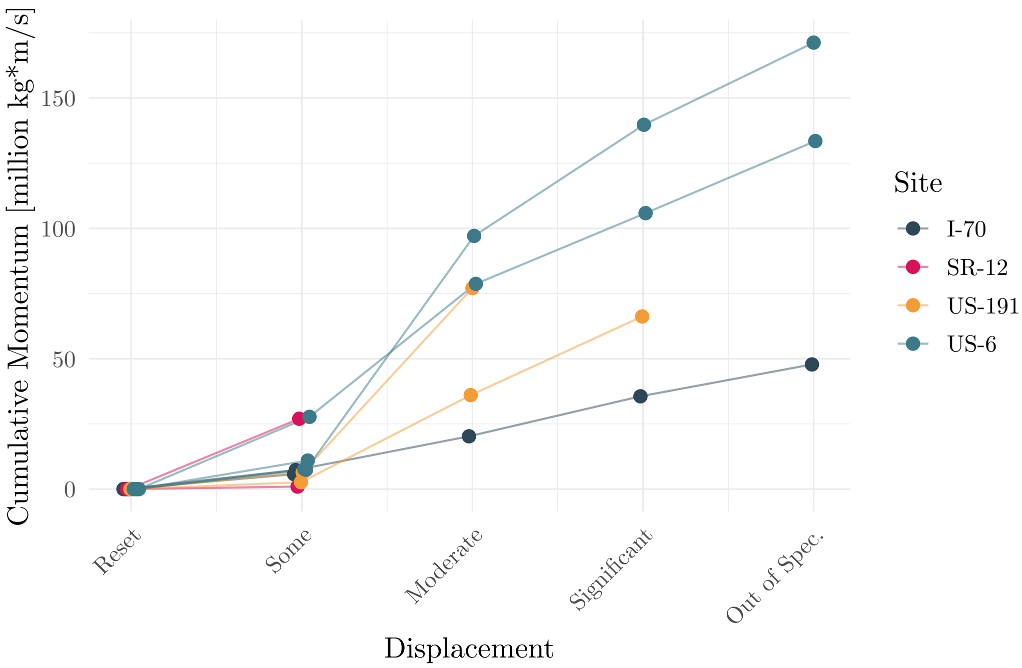
Figure 6 shows the effect of cumulative impact momentum applied to a TPRS array as it transitions between displacement states from a reset to a significant displacement or out of specification condition. Each line in this figure represents a period of observation lasting from when the strip is reset to the baseline condition, to the point where observations ended or workers reset the displaced strips. Lines with higher cumulative momentum represent TPRS that stayed in place longer than lines with lower cumulative momentum. Not all the observed transitions from one state to another could be used: only chains of transitions that started from the initial baseline condition were considered to avoid any confounding variables related to the acceleration of displacement [4]. Figure 6a shows these chains of transitions colored by site, and Figure 6b groups by spacing type; there does not appear to be any strong relationship between spacing and displacement rate, but there is a strong relationship between site and displacement rate. This may be a function of the condition of the strips themselves, which could not be controlled in this research.

3.4.1. Worker Exposure

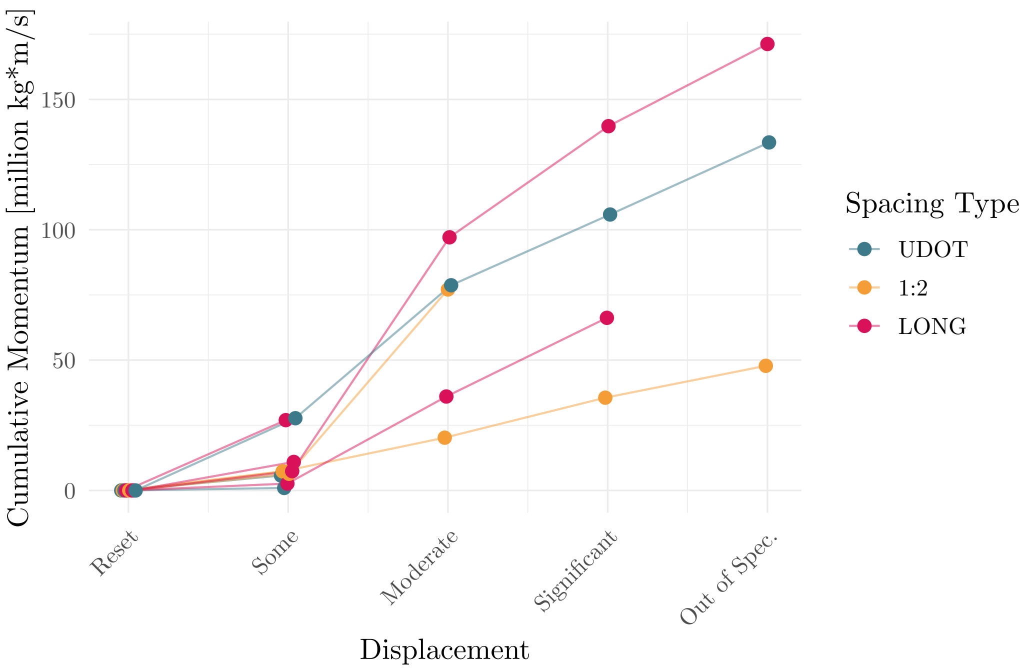
The researchers observed six instances of workers entering the roadway to adjust TPRS. In each case, the workers were able to find a gap in traffic that allowed them to enter the roadway and adjust the strips without incident; the critical gap accepted by the workers was calculated as 33.5 seconds. Figure 7 shows the cumulative distribution of headways for each site, with the critical gap marked as a vertical line. The analysis suggests that at all roadways, a gap of at least 33.5 seconds occurs between at least 15 percent of all headways. On roads with higher volumes and therefore shorter headway distributions, the availability of critical headways may eliminate the possibility for workers to enter the roadway without stopping traffic.

4. Discussion and Limitations

There was no consistent relationship between driver speed and the presence or absence of TPRS, and there was no consistent relationship between driver speeds and the spacing of strips TPRS array. This is contrary to the consensus of the studies in the prior literature, which found TPRS generally slows driving speeds by 3-13 mph. There are several hypotheti-



(a) By site



(b) By spacing

Figure 6. Cumulative impact momentum for transitioning between displacement states.

Table 4. Binomial Models of TPRS Avoidance

	SR-12	US-6	I-70	US-191
(Intercept)	-5.622***	-7.577***	-21.499	-6.343***
	(0.413)	(1.012)	(754.412)	(1.009)
Trucks	0.306	0.536	-0.669+	0.121
	(0.204)	(0.364)	(0.388)	(0.172)
Motorcycles	2.192***	2.874***	3.836***	2.060***
	(0.282)	(0.777)	(0.948)	(0.354)
UDOT Spacing	2.868***	1.875+	16.662	4.313***
	(0.433)	(1.063)	(754.413)	(1.013)
1:2 Spacing	3.092***	3.145**	17.971	4.312***
	(0.423)	(1.043)	(754.412)	(1.012)
Long Spacing	3.165***	2.723**	15.184	4.071***
	(0.424)	(1.030)	(754.413)	(1.016)
Num.Obs.	4963	6324	5933	1796
RMSE	0.229	0.078	0.093	0.284
Log Likelihood	-1003.004	-221.398	-259.041	-505.462

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

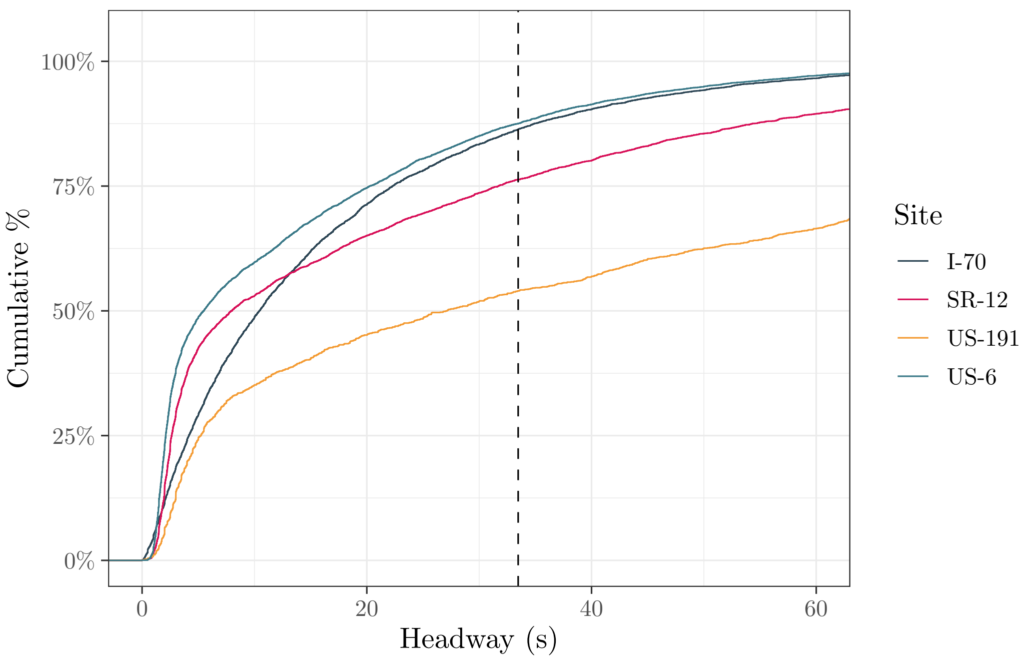


Figure 7. Cumulative distribution of headways at each site, with critical headway marked as a vertical line.

cal explanations for this observed discrepancy. One explanation is that Utah drivers behave differently with respect to TPRS than drivers in the states previously studied. Another is that TPRS technology is more mature now than when first studied; it may be that drivers have become accustomed to TPRS and previous studies observed drivers reacting to a novel technology.

Other explanations for the discrepancy between this study's findings and previous studies in the literature could be based in the methodology used for this research. The observed change in speed between sites was larger than the variance within a site, implying that other variables related to the site may have had a greater impact on drivers' speed than the TPRS treatment. Curves in the road, traffic conditions on any given day, and sight lines from the TPRS array to the work zone could all be factors affecting driver speed. It was not possible for this study to control for these variables as they were endogenous to the site; more sites would need to be observed to control for these elements. The study used commercial radar speed detection units available from UDOT; these units provide only average and 85th percentile speeds within 15-minute time intervals. It may be that TPRS slow only the fastest 15 percent of drivers, a beneficial effect that this study could not have observed. The location of the second radar device was inconsistent between sites — based on the availability of space at the work sites to place the collection device — potentially impacting the measured speed reduction. An ideal location to measure final speed would be at the lane closure, but space was not available at the work sites to place this collection device at that point. A third speed measurement device could also have allowed the researchers to measure the speed of vehicles ahead of TPRS, providing more information on their effectiveness.

Finally, the primary design of TPRS as a traffic control device is to gain the attention of potentially inattentive drivers, with a reduction in speed as a potential consequence. Although change in speed is a common proxy measure for TPRS effectiveness, it would be better to observe driver attention directly. Prior studies have observed this effect by instrumenting the interior of cars with drivers in a closed and controlled setting. Both methods have serious limitations; the most credible approach is probably to use instrumented vehicles on actual roadways, though this presents serious challenges.

In terms of braking response, most drivers were not observed to activate their brake lights at any point as they traversed the TPRS. Of the drivers this study observed activating their brake lights, more did so before striking the TPRS than after. Braking behavior varied more across sites than within sites, indicating strong site-specific effects.

There are multiple interpretations of these findings. If the purpose of TPRS is to gain the attention of distracted drivers, then the fact that more drivers brake before TPRS —- with visual warning rather than tactile warning —- may suggest that visual cues are more influential tactile cues. On the other hand, the fact that some drivers are observed braking after TPRS suggests that for that small number of drivers, TPRS may have played a tactile role in alerting the driver of an upcoming work zone.

A primary limitation to observing brake lights was the difficulty in detecting activated brake lights and the limited observation window before the vehicle encountered TPRS. Some vehicles kept rear-facing tail lights illuminated similar to daytime running lights, relying on an increase in brightness to signal braking. Some brake light housings reflected the sunlight, making it difficult to see whether brake lights were illuminated. Both factors made it difficult to discern activated brake lights from false positives. To mitigate this, Camera 1 was placed closer to the TPRS array to see the vehicles in more detail. However, placing the camera closer shortened the amount of time vehicles were observed before striking the TPRS. Before the TPRS, vehicles were in frame for around 1-2 seconds or less, depending on the speed of traffic. After the TPRS, vehicles were in frame for 3 or more seconds. It is possible that some drivers braked before entering the field of view of Camera 1. The count of vehicles that activated their brake lights after the TPRS is likely more accurate than the counts of vehicles that braked before or not at all. Additional research is needed where a camera is placed farther upstream of the TPRS where drivers first see TPRS. Such footage would provide better insight into how drivers react to TPRS before striking them.

Most drivers in passenger vehicles and trucks do not leave their lane to avoid the TPRS, but a large portion of motorcycle drivers do. Drivers of all vehicle types appeared to avoid the TPRS in a manner similar to passing a slow-moving vehicle; that is, this study did not observe abrupt or sudden evasive maneuvers. The low rates of lane departure amongst drivers of passenger vehicles and trucks suggest that the increased risk of related collisions due to TPRS is low, but additional research would be needed to measure this effect.

TPRS move and displace during operation. The middle strip in a three-strip array appears to displace the most relative to the first and third strips. The amount of displacement was not affected by the spacing between the strips, and there is not a strong relationship between displacement and cumulative traffic momentum. Displacement was more correlated with the site. This could be a function of roadway geometry, pavement condition, or the condition of the strips themselves. All observed sites had asphalt surfaces, but this study did not measure pavement roughness or other specific characteristics that could have explained variation in displacement. This study also did not measure the level of wear of the individual TPRS or their age. Based on the strip designs used at these work zones, it is possible that some strips are more than 10 years old. Additionally, the assumptions made to calculate the cumulative momentum for TPRS displacement are very limited in their precision. While this is the first study to measure TPRS displacement in the field under live traffic loads, additional research is needed to correlate strip age and wear with displacement as well as refine methods to normalize traffic load on TPRS between sites.

Workers enter live traffic streams to correct the installation of displaced strips. This study measured that workers seek headway times in traffic of approximately 34 seconds in order to make these corrections. At all of the sites observed in this research, such a headway was observed at least 20 percent of the time. It may be that at roads with more traffic and therefore smaller headways, workers will accept shorter headways and take more risks. The necessary headway of 34 seconds is calculated from only six observed correction events; more observations may yield a more precise measure.

5. Conclusion

The findings of this research suggest that driver responses to TPRS and the performance of the TPRS themselves are highly variable and strongly dependent on site-specific conditions rather than layout spacing, traffic speeds, or traffic volumes. Changes in vehicle speed showed no consistent relationship with the varied spacing of the TPRS layouts. Likely influences on speed data include uncontrolled differences across work zones and limitations in the setup and equipment used in the field research. Braking and avoidance behaviors were similarly inconsistent and difficult to interpret due to challenges in detecting brake activation and limited visibility. The data suggests that only a minority of drivers respond to TPRS, and of those that do most brake on the approach. Strip displacement also varies widely by site and does not correlate with spacing or estimated traffic momentum, suggesting influence from factors not observed in this study such as pavement condition, age and condition of the individual TPRS devices, and limitations in the method used to estimate momentum. While sufficient headways were present at the observed work zones, the results for worker exposure were based on only six instances when workers were observed to enter the roadway to adjust the TPRS. These limitations underscore the potential of future research implementing more precise measurements, expanded site sampling, and improved methods for assessing driver attention, braking, and TPRS condition.

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